Ordering Multilingual Support Tickets: A Triage Score, and the Waiting-Time Gap It Closes

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***Abstract*—A bank's support desk does not need a label. It needs to know which ticket to open next. Published triage systems predict a per-ticket severity and report classification metrics, leaving the ordering question unanswered. We present a ticket ordering system for a trilingual Sri Lankan retail bank, where customers write in Sinhala, Tamil, and English and render the two local languages in Latin characters as readily as in their native scripts. The Ticket Urgency Score combines a priority posterior, a sentiment posterior, and a per-intent criticality prior estimated from training data, under multiplicative aging, with weights fitted on development data and applied unchanged to test. Evaluated in a discrete-event queue over 200 seeds, it attains 97–99% of the reduction in urgent-ticket waiting time that a perfect classifier would deliver.**

The result that matters for deployment is a fairness one. Under a naive argmax-tier policy, a customer who writes an urgent ticket in romanized Sinhala waits 4.07 minutes longer for a first response than one writing the same ticket in English (95% CI $[2.96, 5.21]$), while the native-script tracks are indistinguishable from English. The comparison is controlled, the romanized track being a deterministic transliteration of the native one. The gap is absent under first-come-first-served and under a gold-label control, so the model causes it rather than the arrival process, and our score cuts it to +0.37 minutes. We also show that classifier accuracy does not transfer to the queue: a 2-point macro-F_1 advantage buys nothing through the median and costs six hours in the tail, because the more accurate model makes fewer errors but 8.5× more confident ones.

***Index Terms*—support-ticket triage, ticket ordering, queueing, scheduling, service fairness, multilingual NLP, code-mixed text, romanization, Sinhala, Tamil**

I. Introduction

A support desk with five agents and a backlog does not consume labels. It consumes an order. Published work on automated ticket triage stops one step short of that: systems predict an escalation risk, a component, or a severity tier and report accuracy or recall against a held-out set [1]–[3]. What happens to the queue once those predictions are used is left to the reader.

The gap matters because the two questions have different answers, and the mapping between them is not monotonic. A classifier that is three points better can produce a queue that is no better at all, because ordering depends on the classifier only where the queue is contended. We show a stronger version of the same point: the more accurate of two classifiers produces the *worse* tail, serving its worst-treated urgent ticket six hours late against the weaker model's ninety-eight minutes.

Sri Lankan retail banking makes this concrete and adds a second dimension. Customers write in Sinhala, Tamil, and English, and they write the two local languages in Latin characters at least as often as in the native abugidas. Those romanized forms have no orthographic standard, and multilingual encoders, pretrained on native-script web text, are measurably worse on them. In a classification table that deficit is an abstraction. Once tickets are ordered it becomes something an operations owner can act on: *customers who write in romanized script wait longer for the same urgent problem.*

That is this paper's central finding, and it is a scheduling property rather than a classification property. Under a naive deployment that takes the argmax of the priority head and serves tier by tier, a Singlish-writing customer with an urgent ticket waits 4.07 minutes longer than an English-writing customer, while native-script Sinhala is indistinguishable from English. Those two tracks hold the same tickets and differ only in orthography, so the comparison is controlled. Four nested controls establish that the model causes it: the gap is absent under first-come-first-served, absent under a gold-label oracle, present only under the predicted labels, and present only on the two romanized tracks.

We then show it is fixable without touching the classifier. The Ticket Urgency Score consumes the same model's output differently, replacing the argmax with the posterior and backstopping it with a script-robust prior over intents, and cuts the gap by an order of magnitude.

Contributions. (1) A ticket ordering score that turns three noisy classifier outputs into one ordering, fitted on development data and applied unchanged to test (Section IV). (2) A queue-level evaluation methodology, and evidence that rank correlation selects the wrong system for a queue (Sections V and VI). (3) A measured cross-script service disparity in minutes of customer waiting time, with four nested controls, and its reduction by an order of magnitude (Section VII). (4) Evidence that classifier accuracy does not transfer to the queue, and a triage-specific error metric that explains why (Section VIII).

II. Related Work

Ticket triage stops at the label. Montgomery et al. [1] model escalation risk over 2.5M IBM support tickets; DeepTriage [2] assigns bug reports to developers; Lee et al. [3] report an industrial incident deployment. All three predict a per-ticket label and report classification metrics, and none asks what the prediction is worth once tickets compete for a finite number of agents. That question is this paper's subject.

Scheduling answers the ordering question. The $c\mu$ rule [4] is the classical result for linear delay cost, with Cobham's priority analysis [5] and Kleinrock's delay-dependent priorities

[6] preceding and extending it, accumulating-priority queues adding aging [7], and the generalised $c\mu$ rule [8] covering convex cost. Our score is best understood as a $c\mu$ rule whose cost term is *learned* rather than assigned: the classical rule assumes a known per-class cost, and ours estimates it per ticket from a posterior. Argon and Ziya [9] give the result that licenses this directly, namely that ordering by a posterior index rather than by its argmax lowers long-run cost, with the gain growing in the variability of the index. We use [10] for objectives and nDCG [11] for queue-head quality.

Fairness here is a scheduling property. Our disparity result concerns groups of *customers* and is denominated in waiting time, which distinguishes it from fairness in ranking [12], where the groups are ranked items and the currency is exposure.

Multilingual and romanized input. The classifiers feeding our queue are multilingual encoders [13]–[15] and Indic-targeted models [16], evaluated on Sinhala and Tamil written in both native and Latin script. Human-typed romanized Tamil [17] is the reference point that makes our synthetically romanized tracks an optimistic bound; Sinhala resources remain thin [18], [19]. Intent labels come from BANKING77 [20]. Because two of our three label columns are model-generated, we report measured ceilings throughout, following the treatment of benchmark label error in [21].

III. The Setting and the Input Layer

A. Corpus

The classifiers feeding the queue are trained on a corpus of 13,077 retail-banking support tickets in five renderings, 65,385 rows. The English track is BANKING77 [20] unmodified, so its 77-way intent column is human gold. Sinhala and Tamil were machine-translated from it, Singlish produced by deterministic rule-based romanization *of the Sinhala track*, and Tanglish translated directly from the English source and romanized. Every ticket keeps its identifier and all three labels across all five tracks.

That provenance decides which pairs are controlled comparisons. **Sinhala and Singlish hold the same content and differ only in script**, so a comparison between them varies orthography alone. **Tamil and Tanglish do not**: they are two independent translations of one English original, not one text in two scripts, so they are not a script pair and we do not read them as one. This also accounts for Tanglish carrying 60.3% out-of-vocabulary tokens against its own training split, against 20–30% for every other track.

Register. Sri Lankan customers writing to a bank in Sinhala routinely keep English banking vocabulary rather than reaching for uncommon native equivalents, and the Sinhala track was hand-corrected to that register. Measured over a 31-term banking lexicon on the test split, it retains 65% of those terms in English against the Tamil track's 6%. We do not offer this as a general contrast between the two languages, since Tamil–English code-mixing is well attested [17], and only the Sinhala track received a hand pass, so register and treatment are confounded here. The consequence we do rely on is narrower: the two native tracks differ in register as well as in language, which is a second reason to treat Sinhala and Singlish as the controlled contrast.

We draw one split at the ticket level and fan it across all five tracks, so a ticket seen in training never appears in test in any rendering: 8,500 / 1,498 / 3,079 tickets, or 42,500 / 7,490 / 15,395 rows. Selection uses development only; test is fitted on training plus development and scored once.

Only intent is human gold. Sentiment and priority were generated by prompting a large language model and carry label noise. Against a 500-ticket human-annotated gold set, sentiment agreement reaches 0.7931 Negative-F_1 and priority 0.7722 macro-F_1; every score below should be read against those ceilings. Sentiment is 93.7% Neutral, so we report Negative-F_1 rather than accuracy. Priority is 53.5 / 36.6 / 9.8 across Low, Medium, and High, so the operationally important class is the rarest.

B. The classifiers, and the deficit that motivates this paper

The queue consumes posteriors from three heads. Priority and sentiment come from a fine-tuned LaBSE encoder [14] reaching 0.8901 macro-F_1 and 0.7138 Negative-F_1 on test; intent comes from the same backbone at 0.8835 macro-F_1. A TF-IDF character n-gram baseline with a linear SVM reaches 0.8706, 0.6653, and 0.8308, and we use it in Section VIII as a substitution control.

The encoder's advantage over that baseline is not spread evenly. On sentiment it gains +0.0834 Negative-F_1 on English and +0.0822 on native-script Sinhala, and an amount indistinguishable from zero on both romanized tracks (Singlish +0.0124, $p = 0.712$; Tanglish +0.0225, $p = 0.490$). Because the Sinhala and Singlish tracks hold the same tickets in the same language and differ only in orthography, this is testable as a difference-in-differences rather than by comparing two p-values [22]: +0.0698, 95% CI $[0.014, 0.128]$, $p = 0.012$. The corresponding Tamil comparison is null ($p = 0.762$), which is the expected result for a pair that does not hold content constant rather than a failure to replicate. The mechanism is tokenizer vocabulary coverage. MuRIL [16] maps 64.5% of Sinhala characters to `[UNK]` and loses 23.4 points on that track alone.

This deficit is the input to everything that follows. Section VII asks what it costs a customer.

IV. The Ticket Urgency Score

A. Definition

For a ticket t evaluated at decision time τ, the score is a static component under multiplicative aging:

$$
\begin{aligned}
U(t,\tau) &= S(t)\left(1 + \alpha\, a(t,\tau)\right) \qquad (1)\\
S(t) &= w_P\, \hat{\mathbb{E}}[\text{sev} \mid t] + w_S\, \hat{P}(\text{Neg} \mid t) + w_I\, \kappa(\hat{\imath}(t))\\
a(t,\tau) &= \left(\tau - \text{arrival}(t)\right)/D
\end{aligned}
$$

with $w \geq 0$, $\sum w = 1$, $\alpha \geq 0$, and D the service window of the ticket's class. Four numbers are fitted. Waiting tickets are served in decreasing U.

Expected severity, not the argmax. $\hat{\mathbb{E}}[\text{sev} \mid t] = \sum_k \hat{p}_k(t)\,\text{sev}_k$ over $\text{sev} = (0, \frac{1}{2}, 1)$. Taking the argmax of the priority head and mapping it to a tier is what a naive deployment does, and it discards exactly what the queue needs: confidence. A confidently-Low ticket and a 0.49/0.51 coin flip both become "Low" and then sit in the same tier in arrival order. Our error analysis finds priority errors concentrate on the Low/Medium boundary, which is precisely where the argmax destroys the most information. Ordering by the posterior rather than its argmax is the property Argon and Ziya [9] prove lowers long-run cost, so this is a theorem rather than a preference.

Sentiment as a nudge, never a gate. At 0.7138 Negative-F_1 the sentiment head misses roughly a third of the negative tickets. A weak signal carrying a small fitted weight inside a score is useful; the same signal as a routing rule would be unsafe.

Learned per-intent criticality. $\kappa(i) = \hat{P}(\text{High} \mid \text{intent} = i)$, 77 values estimated on training and development data only. This is the term that makes (1) a $c\mu$ rule [4] with a *learned* cost rather than an assigned one. It ranks `compromised_card`, `lost_or_stolen_card`, and `card_payment_not_recognised` highest, which is the ordering a bank would choose by hand. It also matters for Section VII: intent is the most accurate of the three heads, so κ acts as a script-robust prior that backstops the priority head where the text is degraded.

Aging is multiplicative, so a stale Low climbs but still climbs more slowly than a stale High, which additive aging cannot express.

B. Fitting

Weights are fitted by grid search on development data over the 3-simplex at 0.05 resolution with a grid on α, then applied unchanged to test: $w = (0.80, 0.10, 0.10)$, $\alpha = 16$. Development posteriors come from a train-only fit, because the saved encoder checkpoints were fitted on training plus development and have memorised it.

The exact vector is not the finding, and we do not present it as one. The surface is flat: 71 of 231 simplex points sit within 5% of the optimum, spanning $w_P \in [0.10, 0.90]$. What the fit establishes is that priority dominates and that no corner of the simplex is competitive, not that 0.80 beats 0.70. We publish the whole surface rather than the argmin. Extending the α grid to 128 confirms the fitted value sits on a plateau, so aging is a policy dial rather than a fitted constant.

An ablation at $\rho = 1.05$ shows the terms are not interchangeable. Priority alone reaches 2.15 minutes mean wait for urgent tickets against the full score's 1.97, and priority with intent reaches 1.85; sentiment on its own degrades the head of the queue. Intent is what earns its place, and it does so through κ rather than as a third severity estimate.

V. Queue-Level Evaluation

Scoring a triage system on classification metrics answers a question the desk did not ask. We evaluate at three levels, each with its own control.

Simulator. A discrete-event M/G/5 queue with log-normal service times (mean 8 minutes, $\sigma = 0.75$), 1,000 tickets per replication, 200 seeds, at utilisation $\rho \in \{0.85, 0.95, 1.05\}$. Service windows are 30, 120, and 480 minutes for High, Medium, and Low. Every one of those is a stated assumption about the desk rather than a measurement from one, which Section IX returns to.

Policies compared. *FIFO* never reads the ticket and is the null. *Tier* takes the argmax of the priority head and serves tier by tier in arrival order, which is what a naive deployment does and is the system our score must beat. *TUS* is (1). *Oracle* orders by gold tier, FIFO within tier, and is a true upper bound: it is the same policy as *tier* with perfect labels, so the pair isolates classifier error, and no system under test can beat it.

Attainment. We report

$$\text{attainment} = \frac{\text{FIFO} - \text{TUS}}{\text{FIFO} - \text{oracle}}$$

which separates two claims that a raw waiting time confounds: whether the scoring function is good, and whether the classifiers are. A score attaining 0.99 of what perfect labels would deliver leaves almost nothing for better classification to buy, and that conclusion is invisible in a classification table.

Statistics. Contrasts are bootstrapped over simulation seeds. Per-track comparisons are reported as *fixed contrasts against English*, never as max-minus-min spread across the five tracks. Spread selects the extreme of five noisy means and is upward-biased: FIFO scores a spread of 16.2 minutes while every one of its fixed contrasts is indistinguishable from zero.

A. An objective that selected a degenerate system

We record one design failure because any triage study can walk into it. Our first fitting objective was absolute tardiness weighted by class severity (0.5/1.0/1.5). Against a 55/36/9 class mix that gives Low 0.275 of the objective's mass against High's 0.143, so it rewarded not starving Low almost twice as much as serving High, and its argmin was $w = (0, 0, 1)$: intent criticality alone, which cannot separate Medium from Low at all. Denominating lateness in units of each class's own service window fixes it. An objective that looks reasonable can select a system that is obviously wrong, and the tell was in the fitted weights rather than in the objective's definition.

VI. Does the Score Order the Queue?

A. Attainment against a perfect classifier

Table I reports mean time-to-first-response for gold-High tickets. FIFO degrades sharply with load, as a queue with no triage must. The score tracks the oracle closely at every load, attaining 97.1% of the achievable improvement at $\rho = 0.85$ and 99.0% at $\rho = 1.05$.

The remaining headroom is about a tenth of a minute. Within this workload, **further classifier accuracy cannot buy much**, because the gap between our noisy labels and perfect ones is nearly closed already by consuming the noisy labels properly. Section VIII tests that directly by substituting a weaker classifier.

TABLE I
MEAN WAIT FOR GOLD-HIGH TICKETS, MINUTES, 200 SEEDS. ATTAINMENT IS (FIFO − TUS)/(FIFO − oracle).

ρ	FIFO	TUS	Oracle	Attainment
0.85	5.78	1.13	0.99	**0.971** [0.966, 0.977]
0.95	16.76	1.54	1.30	**0.985** [0.982, 0.987]
1.05	52.46	1.98	1.47	**0.990** [0.988, 0.992]

TABLE II
ORDERING QUALITY ON TEST, ORDERED ONCE. THE ARGMAX-TIER POLICY HAS THE BEST RANK CORRELATION AND THE WORST QUEUE HEAD.

System	Kendall τ_b	nDCG@10	P(High)@10
Predicted tier (argmax)	**0.822**	0.351	0.50
TUS (LaBSE)	0.686	**1.000**	**1.00**
Priority posterior only	0.690	**1.000**	**1.00**
Oracle (gold tier)	1.000	**1.000**	**1.00**

TABLE III
MEAN WAIT FOR GOLD-HIGH TICKETS, EACH TRACK MINUS ENGLISH, MINUTES, $\rho = 1.05$, 200 SEEDS, BOOTSTRAP CIS. READ AS FOUR NESTED CONTROLS.

Policy	singlish	sinhala	tamil	tamilish
FIFO	−0.85	−0.22	−0.23	−0.82
Tier	**+4.07**	+0.37	−0.01	**+6.99**
TUS	**+0.37**	+0.24	+0.02	**+0.50**
Oracle	−0.02	0.00	0.00	−0.01

B. Rank correlation selects the wrong system

The obvious way to score an ordering is a rank correlation against the gold ordering. On our data it is actively misleading (Table II).

The argmax-tier policy has the highest τ_b of any system and the worst queue head: half of the first ten tickets an agent opens are not urgent, against a base rate of 9.5%. The explanation is structural. Kendall's τ rewards getting the global ordering right, and a 3-valued tier score matches a 3-valued gold label well. But within the top tier every ticket ties, so the order an agent actually experiences is arbitrary.

A queue is consumed from the top, and τ averages over the whole list. Selecting on it would have picked exactly the wrong system. This is the concrete justification for evaluating triage in queue terms rather than with a ranking correlation, and it is why we lead with Precision@k over gold-High, which an operations owner can read without a statistics background.

VII. SCRIPT-BASED SERVICE DISPARITY

This is the finding the system exists to expose. A classification deficit on romanized text, which Section III measures as a difference-in-differences of +0.0698, is an abstraction. Expressed as waiting time it becomes something an operations owner can act on.

Table III reports mean wait for gold-High tickets, each track minus English, at $\rho = 1.05$ over 200 seeds. Bold marks intervals excluding zero.

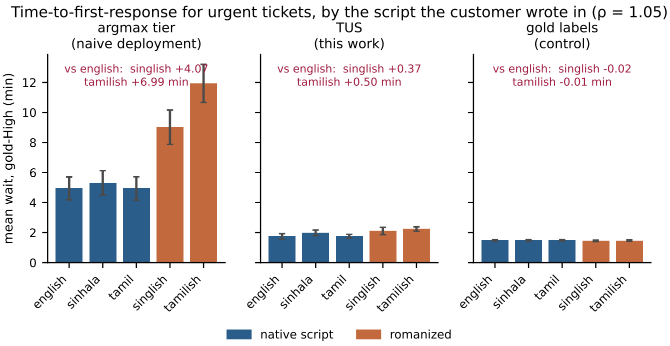

Fig. 1. Mean wait for urgent tickets by track, relative to English. The gap appears only under predicted labels and only on the romanized tracks, and the score removes most of it.

Under FIFO, nothing. All four contrasts straddle zero. FIFO never reads the ticket, so the arrival process contributes no gap.

Under the gold-label control, nothing. Same tickets, same arrival stream, labels not predicted. Any gap appearing between these two rows is therefore attributable to the model.

Under the naive deployment, exactly and only the two romanized tracks. Singlish +4.07 minutes (CI $[2.96, 5.21]$) and Tanglish +6.99 (CI $[5.76, 8.19]$), while native-script Sinhala and Tamil are indistinguishable from English. A customer who writes their urgent problem in Latin-script Tamil waits seven minutes longer for a first response than a customer who writes the same problem in English. This is sharper than the classification result it derives from, because the two native tracks come out at zero rather than merely smaller.

The score cuts it by an order of magnitude. Singlish falls from +4.07 to +0.37 and Tanglish from +6.99 to +0.50. Paired on the simulation seed, the score removes 3.70 minutes (CI $[2.57, 4.85]$) of the Singlish gap and 6.49 (CI $[5.32, 7.77]$) of the Tanglish one. It does not remove all of it, and we do not claim otherwise: both residuals still exclude zero. Fig. 1 shows the four policies together.

Why the score helps. Two mechanisms, both by design. The continuous posterior avoids the tier collapse that makes the argmax policy discriminate hardest exactly where it is least accurate: a degraded romanized ticket produces a hedged posterior, which the argmax flattens into a confident tier and the score preserves as a mid-ranking. And κ(intent) is a script-robust prior, because intent is the most accurate of the three heads, so it backstops the priority head where the text is degraded.

The finding rests on Singlish alone, by design. Tanglish is a separate translation from the English source rather than a romanization of the Tamil track (Section III), and it carries 60.3% out-of-vocabulary tokens against its own training split, so its +6.99 is inflated by input degradation as well as by script and we treat it as corroboration rather than evidence. Singlish carries the result on its own: +4.07 minutes, an ordinary out-of-vocabulary profile, and the same content as the Sinhala track that shows no gap at all. The controlled comparison and

TABLE IV
SAME SCORE, SAME WEIGHTS, POSTERIORS SWAPPED. $\rho = 1.05$, 200 SEEDS, WAITS IN MINUTES FOR GOLD-HIGH TICKETS.

	LaBSE	TF-IDF
priority macro-F_1 (test)	**0.8901**	0.8706
mean wait	1.98	**1.80**
median wait	1.11	**1.10**
p95 wait	**5.32**	5.42
worst-case wait	362.1	**97.9**
gold-High missed	**165**	193
of which buried below 0.1	68	**8**

the significant one are the same pair.

VIII. Accuracy Does Not Transfer to the Queue

Section VI found little headroom above the score. We test that directly by holding the score and its weights fixed and swapping the classifier underneath it, replacing the LaBSE priority and sentiment posteriors with those of the TF-IDF baseline (Table IV).

Through the median and the 95th percentile the two are indistinguishable, at tenths of a minute and with the sign changing between rows. They separate in the tail, where the more accurate model's worst-served urgent ticket waits **six hours against the weaker model's ninety-eight minutes**.

The claim is not that TF-IDF wins. It is that a two-point macro-F_1 advantage buys nothing through the body of the queue and costs something at the extreme, which no classification table would report.

The mechanism is a metric F_1 cannot see. LaBSE misses fewer urgent tickets than TF-IDF, 165 against 193, and buries far more of the ones it misses: 68 fall below a score of 0.1 against TF-IDF's 8. Fewer mistakes, 8.5× more confident ones. A confidently-missed urgent ticket falls to the bottom of the queue and stays there, which is the six-hour worst case above, while a hedged miss lands mid-queue and is picked up soon. Macro-F_1 counts both as one error.

We therefore propose **buried urgent tickets**, the count of gold-High tickets scored below a low threshold, as a triage-specific error metric worth reporting beside F_1. It is what the queue actually feels.

A. A hypothesis we refuted

We expected calibration to explain the disparity and to fix it cheaply. The diagnosis supported it: LaBSE is more accurate than TF-IDF, 3.7× worse calibrated (ECE 0.0658 against 0.0180), and worst calibrated on exactly the track it is least accurate on (Tanglish, 0.1071). That is a double penalty, and temperature scaling [23] is the standard remedy.

Fitting the temperature on a clean held-out development set fixed the calibration and made the queue *worse*. Mean wait for urgent tickets rose from 1.98 to 2.65 minutes and the Tanglish-minus-English gap widened from +0.50 to +0.97. The reason is that expected calibration error measures agreement between top-1 confidence and accuracy, while a queue needs posteriors that *separate* tickets, and temperature scaling compresses exactly that separation. "Calibrate before you score" was wrong here, and we report it because the diagnosis is convincing enough that others will try it.

IX. Discussion and Limitations

A. What a support desk should take from this

Consume the posterior, not the argmax. The single largest effect we measure comes from changing how an existing model's output is read, at no training cost: the argmax-tier policy creates a seven-minute cross-script disparity that the same model's posterior largely removes.

Audit triage fairness in waiting time, not in F_1. A per-group accuracy table would have shown a few points of deficit on romanized tracks and given no sense of what it costs a customer. The same deficit, denominated in minutes and controlled against FIFO and a gold-label oracle, names both a cost and a cause.

Better classification is not the next thing to buy. Our score attains 99% of what perfect labels would deliver, and substituting a weaker classifier leaves the median untouched. Effort is better spent on how predictions are consumed, and on the tail behaviour that F_1 hides.

B. Limitations

The simulation parameters are assumed. Arrival rate, service distribution, agent count, and the service windows are stated assumptions rather than measurements from a live desk. This is the largest threat to the paper. We mitigate it by sweeping utilisation and by reporting attainment, which is a ratio and so is less sensitive to the absolute scale than raw waiting times, but a parameterisation drawn from published helpdesk statistics would be stronger.

The romanized tracks are synthetic, generated from the native tracks rather than typed by people [17]. Real romanized input is harder, so the disparity we measure is a conservative estimate.

The Tanglish rendering is defective, at 60.3% out-of-vocabulary against its own training split. Singlish carries the disparity result independently.

Two label columns are model-generated, with ceilings of 0.7931 and 0.7722 reported throughout, and the sentiment ceiling rests on one annotator and 31 negative tickets.

Confidence intervals are within-fit. Classification intervals resample the test set rather than the training run, and one seed repeat moved a sentiment headline by 0.0158. Queue intervals resample simulation seeds and are not subject to this.

The weight vector is not identified. The development surface is flat across $w_P \in [0.10, 0.90]$; we claim only that priority dominates and that no corner is competitive.

X. Conclusion

We built a ticket ordering system for a trilingual bank support desk and evaluated it where it is used, in a queue, rather than where it is convenient to measure, in a classification table. The Ticket Urgency Score combines a priority posterior, a sentiment posterior, and a learned per-intent criticality under multiplicative aging, and attains 97–99% of the reduction in urgent-ticket waiting time that a perfect classifier would deliver.

The finding we would put in front of an operations owner is that triage fairness is a scheduling property. A naive argmax deployment makes customers who write romanized Sinhala or Tamil wait 4.07 and 6.99 minutes longer for the same urgent problem, with the gap absent under first-come-first-served, absent under a gold-label control, and absent on both native-script tracks. Reading the same model's posterior through our score cuts it to +0.37 and +0.50.

Two negative results travel with it. Rank correlation selects the worst system in our comparison, because a queue is consumed from the top and τ averages over the whole list. And classifier accuracy does not transfer: a two-point macro-F_1 advantage leaves the median wait unchanged and produces a six-hour tail, because the stronger model makes fewer errors but 8.5× more confident ones.

The corpus, the frozen split, and the model checkpoints are publicly available under CC BY 4.0 with attribution to BANKING77.

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Portions of the prose in Sections I–X were drafted with the assistance of a large language model (Anthropic Claude, Opus 5) used at the level of drafting and copy-editing from author-supplied experimental results, outlines, and notes. The system generated no experimental result, number, table, or figure. All experiments, analyses, and conclusions are the authors' own; every reported value is produced by a script in the released code and was verified against its generating artifact. The authors reviewed and edited all such text and take full responsibility for the content of this article.

References

[1] L. Montgomery, D. Damian, T. Bulmer, and S. Quader, "Customer support ticket escalation prediction using feature engineering," *Requirements Engineering*, vol. 23, no. 3, pp. 333–355, 2018.

[2] S. Mani, A. Sankaran, and R. Aralikatte, "DeepTriage: Exploring the effectiveness of deep learning for bug triaging," in *Proceedings of the ACM India Joint International Conference on Data Science and Management of Data (CoDS-COMAD)*. Association for Computing Machinery, 2019, pp. 171–179.

[3] S.-R. Lee, M.-J. Heo, C.-G. Lee, M. Kim, and G. Jeong, "Applying deep learning based automatic bug triager to industrial projects," in *Proceedings of the 2017 11th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*. Association for Computing Machinery, 2017, pp. 926–931.

[4] D. R. Cox and W. L. Smith, *Queues*. London: Methuen / Chapman and Hall, 1961.

[5] A. Cobham, "Priority assignment in waiting line problems," *Journal of the Operations Research Society of America*, vol. 2, no. 1, pp. 70–76, 1954.

[6] L. Kleinrock, "A delay dependent queue discipline," *Naval Research Logistics Quarterly*, vol. 11, no. 3-4, pp. 329–341, 1964.

[7] D. A. Stanford, P. Taylor, and I. Ziedins, "Waiting time distributions in the accumulating priority queue," *Queueing Systems*, vol. 77, no. 3, pp. 297–330, 2014.

[8] J. A. van Mieghem, "Dynamic scheduling with convex delay costs: The generalized cμ rule," *The Annals of Applied Probability*, vol. 5, no. 3, pp. 809–833, 1995.

[9] N. T. Argon and S. Ziya, "Priority assignment under imperfect information on customer type identities," *Manufacturing & Service Operations Management*, vol. 11, no. 4, pp. 674–693, 2009.

[10] M. L. Pinedo, *Scheduling: Theory, Algorithms, and Systems*, 6th ed. Springer, 2022.

[11] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002.

[12] A. Singh and T. Joachims, "Fairness of exposure in rankings," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. Association for Computing Machinery, 2018, pp. 2219–2228.

[13] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, "Unsupervised cross-lingual representation learning at scale," in *Proc. 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2020, pp. 8440–8451. [Online]. Available: https://aclanthology.org/2020.acl-main.747/

[14] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "Language-agnostic BERT sentence embedding," in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2022, pp. 878–891. [Online]. Available: https://aclanthology.org/2022.acl-long.62/

[15] M. Marone, O. Weller, W. Fleshman, E. Yang, D. Lawrie, and B. Van Durme, "mmbert: A modern multilingual encoder with annealed language learning," 2025. [Online]. Available: https://arxiv.org/abs/2509.06888

[16] S. Khanuja, D. Bansal, S. Mehtani, S. Khosla, A. Dey, B. Gopalan, D. K. Margam, P. Aggarwal, R. T. Nagipogu, S. Dave, S. Gupta, S. C. B. Gali, V. Subramanian, and P. Talukdar, "MuRIL: Multilingual representations for indian languages," 2021. [Online]. Available: https://arxiv.org/abs/2103.10730

[17] B. R. Chakravarthi, V. Muralidaran, R. Priyadharshini, and J. P. McCrae, "Corpus creation for sentiment analysis in code-mixed Tamil-English text," in *Proceedings of the 1st Joint Workshop on Spoken Language Technologies for Under-resourced languages (SLTU) and Collaboration and Computing for Under-Resourced Languages (CCURL)*. Marseille, France: European Language Resources Association, 2020, pp. 202–210. [Online]. Available: https://aclanthology.org/2020.sltu-1.28/

[18] N. de Silva, "Survey on publicly available sinhala natural language processing tools and research," 2019.

[19] D. Sumanathilaka, S. Perera, S. Dharmasiri, M. Athukorala, A. D. Herath, R. Dias, P. Gamage, R. Weerasinghe, and Y. H. P. P. Priyadarshana, "Swa-bhasha resource hub: Romanized sinhala to sinhala transliteration systems and data resources," 2026.

[20] I. Casanueva, T. Temčinas, D. Gerz, M. Henderson, and I. Vulić, "Efficient intent detection with dual sentence encoders," in *Proc. 2nd Workshop on Natural Language Processing for Conversational AI*. Association for Computational Linguistics, 2020, pp. 38–45. [Online]. Available: https://aclanthology.org/2020.nlp4convai-1.5/

[21] C. G. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. [Online]. Available: https://arxiv.org/abs/2103.14749

[22] A. Gelman and H. Stern, "The difference between "significant" and "not significant" is not itself statistically significant," *The American Statistician*, vol. 60, no. 4, pp. 328–331, 2006.

[23] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html